Exercises 7.2 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29 (another answer: $\frac{1}{8}\sec^8 x - \frac{1}{6}\sec^6 x + C$), 33, 39, 41, 45, 47, 61, 63.

5. $\int \sin^5(2t)\cos^2(2t)\,dt$

$= \int \sin(2t)\cdot[\sin^2(2t)]^2\cos^2(2t)\,dt$

$= \int \sin(2t)\cdot[1-\cos^2(2t)]^2\cos^2(2t)\,dt$

Let $u = \cos(2t)$; $\frac{du}{dx} = -2\sin(2t) \Rightarrow dx = \frac{du}{-2\sin(2t)}$

$= \int [1-u^2]^2 u^2 \cdot \frac{du}{-2}$

$= -\frac{1}{2}\int (1-2u^2+u^4)u^2\,du$

$= -\frac{1}{2}\int u^2 - 2u^4 + u^6\,du$

$= -\frac{1}{2}\left(\frac{u^3}{3} - \frac{2u^5}{5} + \frac{u^7}{7}\right) + C$

$= -\frac{u^3}{6} + \frac{u^5}{5} - \frac{u^7}{14} + C$

$= -\frac{\cos^3(2t)}{6} + \frac{\cos^5(2t)}{5} - \frac{\cos^7(2t)}{14} + C$

7. $\int_0^{\pi/2} \cos^2\theta \, d\theta$

$= \int_0^{\frac{\pi}{2}} \frac{\cos 2\theta + 1}{2} \, d\theta$

$= \frac{1}{2} \int_0^{\frac{\pi}{2}} \cos 2\theta + 1 \, d\theta$

$= \frac{1}{2} \left[\frac{\sin 2\theta}{2} + \theta \right]_0^{\frac{\pi}{2}}$

$= \frac{1}{2} \left[\left(\frac{0}{2} + \frac{\pi}{2} \right) - (0 + 0) \right]$

$= \frac{\pi}{4}$

9. $\int_0^{\pi} \cos^4(2t)\, dt$

$= \int_0^{\pi} [\cos^2(2t)]^2 \, dt$

$= \int_0^{\pi} \left(\frac{\cos 4t + 1}{2}\right)^2 dt$

$= \frac{1}{2} \int_0^{\pi} \cos 4t + 1 \, dt$

$= \frac{1}{2} \left[\frac{\sin 4\pi}{4} + t \right]_0^{\pi}$

$= \frac{1}{2} [(0 + \pi) - (0 + 0)]$

$= \frac{\pi}{2}$

11. $\int_0^{\pi/2} \sin^2 x \cos^2 x \, dx$

$= \int_0^{\frac{\pi}{2}} (\sin x \cos x)^2 \, dx$

$= \int_0^{\frac{\pi}{2}} \left(\frac{\sin 2x}{2}\right)^2 dx$

$= \frac{1}{4} \int_0^{\frac{\pi}{2}} \sin^2 2x \, dx$

$= \frac{1}{4} \int_0^{\frac{\pi}{2}} \frac{1-\cos 4x}{2} \, dx$

$= \frac{1}{8} \int_0^{\frac{\pi}{2}} 1 - \cos 4x \, dx$

$= \frac{1}{8} \left[x - \frac{\sin 4x}{4}\right]_0^{\frac{\pi}{2}}$

$= \frac{1}{8} \left[\left(\frac{\pi}{2} - 0\right) - 0\right]$

$= \frac{\pi}{16}$

$\cos 2x = 1 - 2\sin^2 x$

$\sin^2 x = \frac{1-\cos 2x}{2}$

$\cos 2x \cdot 2$

13. $\int \sqrt{\cos\theta}\,\sin^3\theta\,d\theta$

$= \int \sqrt{\cos\theta}\,\sin^2\theta\cdot\sin\theta\,d\theta$

$= \int \sqrt{\cos\theta}\,(1-\cos^2\theta)\sin\theta\,d\theta$

let $u=\cos\theta$; $\frac{du}{dx} = -\sin x \Rightarrow dx = \frac{du}{-\sin x}$

$= \int \sqrt{u}\cdot(1-u^2)\sin\theta\cdot\frac{du}{-\sin\theta}$

$= -\int \sqrt{u}\,(1-u^2)\,du$

$= -\int u^{\frac{1}{2}} - u^{\frac{5}{2}}\,du$ $= -\left(\frac{2}{3}u^{\frac{3}{2}} - \frac{u^{\frac{7}{2}}}{\frac{7}{2}}\right)$

$= -\frac{2}{3}u^{\frac{3}{2}} + \frac{u^{\frac{7}{2}}}{\frac{7}{2}}$

$= -\frac{2}{3}\cdot(\cos\theta)^{\frac{3}{2}} + \frac{\cos^3\theta}{\frac{7}{2}} + \dots$

$\sqrt{u}\cdot u^2$

$u^{\frac{1}{2}}\cdot u^2$

$u^{\frac{5}{2}}$

15. $\int \sin x \sec^5 x \, dx$

$= \int \frac{\sin x}{\cos^5 x} \, dx$

let $u = \cos x$; $\frac{du}{dx} = -\sin x \Rightarrow dx = \frac{du}{-\sin x}$

$= \int \frac{\sin x}{u^5} \cdot \frac{du}{-\sin x}$

$= -\int \frac{1}{u^5} \, du$

$= \frac{u^{-4}}{-4} + C$

$= -\frac{1}{4} \frac{1}{\cos^4 x} + C$

17. $\int \cot x \cos^2 x \, dx$

$= \int \frac{\cos x}{\sin x} (1 - \sin^2 x) \, dx$

$= \int \frac{(1-\sin^2 x}{\sin x} \cdot \cos x \, dx$

Let $u = \sin x$; $\frac{du}{dx} = \cos x \Rightarrow dx = \frac{1}{\cos x}$

$= \int \frac{1-u^2}{u} \cdot du$

$= \int u^{-1} - u \, du = \ln|u| - \frac{u^2}{2} + C$

$= \ln|\sin x| - \frac{\sin^2 y}{2} + C$

19. $\int \sin^2 x \sin 2x \, dx$

$= \int \sin^2 x \cdot 2\sin x \cos x \, dx$

$= \int (1 - \cos^2 x)\, 2\sin x \cos x \, dx$

let $u = \cos x$; $\frac{du}{dx} = -\sin x \Rightarrow dx = \frac{du}{-\sin x}$

$= \int (1 - u^2)\, 2u \cdot \sin x \cdot \frac{du}{-\sin x}$

$= -2\int (1 - u^2)\, u \, du$

$= -2\int u - u^3 \, du = -2\left(\frac{u^2}{2} - \frac{u^4}{4}\right) + C$

$= -u^2 + \frac{u^4}{2} + C$

$= -\cos^2 x + \frac{\cos^4 x}{2} + C$

u

$= 2\int \sin^3 x \cos x \, dx$

let $u = \sin x$; $dx = \frac{du}{\cos x}$

$= 2\int u^3 \, du$

$= \frac{1}{2} u^4 + C$

$= \frac{1}{2} \sin^4 x + C$

21. $\int \tan x \sec^3 x \, dx$

$\tan x$ $\quad \sec^2 x$

$\sec x$ $\quad \sec x \tan x$

$= \int \sec x \tan x \cdot \sec^2 x \, dx$

Let $u = \sec x$; $\frac{du}{dx} = \sec x \tan x \Rightarrow dx = \frac{du}{\sec x \tan x}$

$= \int u^2 \cdot du = \frac{u^3}{3} + C$

$= \frac{\sec^3 x}{3} + C$

23. $\int \tan^2 x \, dx$

$= \int \sec^2 x - 1 \, dx$

$= \tan x - x + C$

$\sin^2 x + \cos^2 x = 1$

$1 + \cot^2 x = \csc^2 x$

$\tan^2 x + 1 = \sec^2 x$

25. $\int \tan^4 x \sec^6 x \, dx$

$= \int \tan^4 x \sec^4 x \cdot \sec^2 x \, dx$

$= \int \tan^4 x (\tan^2 x + 1)^2 \cdot \sec^2 x \, dx$

let $u = \tan x$; $\frac{du}{dx} = \sec^2 x \Rightarrow dx = \frac{du}{\sec^2 x}$

$= \int u^4 (u^2 + 1)^2 \cdot du$

$= \int u^4 (u^4 + 2u^2 + 1) \, du$

$= \int u^8 + 2u^6 + u^4 \, du$

$= \frac{u^9}{9} + \frac{2u^7}{7} + \frac{u^5}{5} + C$

$= \frac{\tan^9 x}{9} + \frac{2\tan^7 x}{7} + \frac{\tan^5 x}{5} + C$

27. $\int \tan^3 x \sec x \, dx$

$= \int \tan^2 x \cdot \sec x \cdot \tan x \, dx$

$= \int (\sec^2 x - 1) \sec x \tan x \, dx$

Let $u = \sec x$; $\frac{du}{dx} = \sec x \tan x$ $\Rightarrow dx = \frac{du}{\sec x \tan x}$

$= \int (u^2 - 1) \cdot du$

$= \frac{u^3}{3} - u + C$

$= \frac{\sec^3 x}{3} - \sec x + C$

29. $\int \tan^3 x \sec^6 x \, dx$

$= \int \tan^3 x \cdot \sec^4 x \cdot \sec^2 x \, dx$

$= \int \tan^3 x \cdot (\sec^2 x)^2 \cdot \sec^2 x \, dx$

$= \int \tan^3 x (\tan^2 x + 1)^2 \sec^2 x \, dx$

let $u = \tan x$; $\frac{du}{dx} = \sec^2 x \Rightarrow dx = \frac{du}{\sec^2 x}$

$= \int u^3 (u^2+1)^2 \, du$

$= \int u^3 (u^4 + 2u^2 + 1) \, du = \int u^7 + 2u^5 + u^3 \, du$

$= \frac{u^8}{8} + \frac{u^6}{3} + \frac{u^4}{4} + C$

$= \frac{\tan^8 x}{8} + \frac{\tan^6 x}{3} + \frac{\tan^4 x}{4} + C$

33. $\int \frac{1-\tan^2 x}{\sec^2 x}\,dx$

$= \int \frac{1-(\sec^2 x - 1)}{\sec^2 x}\,dx$

$= \int \frac{2-\sec^2 x}{\sec^2 x}\,dx$

$= -x + C$

$= \int \left(1 - \frac{\sin^2 x}{\cos^2 x}\right)\cdot \cos^2 x\,dx$

$= \int \cos^2 x - \sin^2 x\,dx$

$= \int \cos 2x\,dx = \frac{\sin 2x}{2} + C$

39. $\int_{\pi/4}^{\pi/2} \cot^5\phi \csc^3\phi \, d\phi$

when $x=\frac{\pi}{2}$, $u=1$

when $x=\frac{\pi}{4}$, $u=\sqrt{2}$

$= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot\phi \cdot \cot^4\phi \cdot \csc^3\phi \, d\phi$

$= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot\phi \, (\csc^2 x - 1)^2 \csc^3\phi \, d\phi$

let $u = \csc x \Rightarrow \frac{du}{dx} = -\csc x \cot x \Rightarrow dx = \frac{du}{-\csc x \cot x}$

$= -\int_{\sqrt{2}}^{1} (u^2-1)^2 u^2 \, du$

$= -\int_{\sqrt{2}}^{1} (u^4 - 2u^2 + 1) u^2 \, du$

$= -\int_{\sqrt{2}}^{1} (u^6 - 2u^4 + u^2) \, du$

$= -\left[\frac{u^7}{7} - \frac{2u^5}{5} + \frac{u^3}{3}\right]_{\sqrt{2}}^{1}$

$= -\left[\frac{\csc^7 x}{7} - \frac{2\csc^5 x}{5} + \frac{\csc^3 x}{3}\right]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$

$= -\left[\left(\frac{1}{7\sin^7(\frac{\pi}{2})} - \frac{2}{5\sin^5(\frac{\pi}{2})} + \frac{1}{3\sin^3(\frac{\pi}{2})}\right) - \left(\frac{1}{7\sin^7(\frac{\pi}{4})} - \frac{2}{5\sin^5(\frac{\pi}{4})} + \frac{1}{3\sin^3(\frac{\pi}{4})}\right)\right]$

$= \left[\frac{1}{7} - \frac{2}{5} + \frac{1}{3} - \left(\frac{1}{7\times(\frac{\sqrt{2}}{2})^7} - \frac{2}{5\times(\frac{\sqrt{2}}{2})^5} + \frac{1}{3\times(\frac{\sqrt{2}}{2})^3}\right)\right]$

$= -0.22$

41. $\int \csc x \, dx$

$$= \int \csc x \cdot \frac{\csc x}{\csc x} \, dx$$

$$= \int \frac{\csc^2 x}{\csc x} \, dx$$

$$= \int \csc^2 x \cdot \sin x \, dx$$

$$= -\cot x \cdot \cos x + C$$

$\tan x$ $\sec^2 x$

$\sec x$ $\sec x \tan x$

$\cot$ $-\csc^2 x$

$\csc$ $-\csc x \cot x$

$$= \int \csc x \cdot \frac{\csc x + \cot x}{\csc x + \cot x} \, dx$$

$$= \int \frac{\csc^2 x + \csc x \cot x}{\csc x + \cot x} \, dx$$

let $u = \csc x + \cot x$; $dx = -\dfrac{du}{\csc x \cot x + \csc^2 x}$

$$= -\int \frac{1}{u} \, du = -\ln|\csc x + \cot x| + C$$

45. $\int_0^{\pi/2} \cos 5t \cos 10t \, dt$

(a) $\sin mx \cos nx = \frac{1}{2}[\sin(m+n)x + \sin(m-n)x]$

(b) $\sin mx \sin nx = -\frac{1}{2}[\cos(m+n)x - \cos(m-n)x]$

(c) $\cos mx \cos nx = \frac{1}{2}[\cos(m+n)x + \cos(m-n)x]$

$= \int_0^{\frac{\pi}{2}} \cos 5t \,(2\cos^2 5t - 1)\, dt$

$= \int_0^{\frac{\pi}{2}} 2\cos^3 5t - \cos 5t \, dt$

$= \int_0^{\frac{\pi}{2}} (\cos 10t + 1)\cos 5t - \cos 5t \, dt$

$= \int_0^{\frac{\pi}{2}} \cos 10t \cos 5t \, dt$

$= \int_0^{\frac{\pi}{2}} \frac{\cos^2 10t + 1}{2} \, dt = \int_0^{\frac{\pi}{2}} \frac{\frac{\cos 20t + 1}{2} + 1}{2} \, dt$

$= \frac{1}{4}\int_0^{\frac{\pi}{2}} \cos 20t + 3 \, dt$

$= \frac{1}{4}\left[\frac{\sin 20t}{20} + 3t\right]_0^{\frac{\pi}{2}}$

$= \frac{1}{4}\left[\left(0 + \frac{3\pi}{2}\right) - 0\right]$

$= \frac{3\pi}{8}$

45. $\int_0^{\pi/2} \cos 5t \cos 10t \, dt$

$= \int_0^{\frac{\pi}{2}} \frac{1}{2}[\cos(5t-10t) + \cos(5t+10t)]\, dt$

$= \frac{1}{2} \int_0^{\frac{\pi}{2}} \cos(-5t) + \cos(15t)$

47. $\int \frac{\sin^2(1/t)}{t^2}\,dt$

let $u = \frac{1}{t}$, $\frac{du}{dt} = -\frac{1}{t^2} \Rightarrow dt = -t^2\,du$

$= \int \frac{\sin^2(u)}{t^2} \cdot (-t^2)\,du$

$= -\int \sin^2(u)\,du = -\int \frac{1-\cos 2u}{2}\,du$

$= -\frac{u}{2} + \frac{1}{2}\int \cos 2u\,du$

$= -\frac{u}{2} + \frac{\sin 2u}{4} + C$

$= -\frac{1}{2t} + \frac{\sin(\frac{2}{t})}{4} + C$